

From the Lab to the Real World: Building Usable Socially Assistive Robots

SOOMIN SHIN

Personal Website ◇ Google Scholar ◇ Github

Email: soomin.shin@uwaterloo.ca

Ph.D. candidate specializing in **Human-Robot Interaction (HRI)** and **LLM-driven Social Robots**, focused on building readily deployable systems for real-world settings that can be used by non-technical users. Expert at designing, prototyping, and deploying socially assistive robots with advanced interaction capabilities. Passionate about translating research into accessible, field-ready systems, with a strong publication record in top-tier venues and hands-on experience in collaborative, interdisciplinary environments.

EDUCATION

University of Waterloo - *PhD. in Electrical and Computer Engineering*
Supervisor: Prof. Kerstin Dautenhahn

Present - 09/2027 Expected

Korea University - *MSc. in Brain and Cognitive Engineering*

Seoul Women's University - *B.E. in Multimedia, B.A. in Child Studies*

RESEARCH EXPERIENCE

Graduate Researcher

Social and Intelligent Robotics Research Lab, University of Waterloo

- **Research Focus:** Developing social robots for the real clinical setting **shin2025co_design**, [1]
- Designed and implemented a system for a **general-purpose therapeutic robot**, engineering an architecture that integrates LLMs with a robotic platform using PyTorch and ROS.
- Generated expressive, goal-oriented robot gestures using the MoveIt Motion Planning Framework and inverse kinematics, improving the robot's non-verbal communication.
- Developed a scalable, deployable prototype by translating ambiguous user needs into a concrete system design, leading a two-year development cycle in collaboration with a team of clinical experts.
- Developed a **physical robotic system within a clinical setting**, managing the full-stack development of the robot software and a Flask-based web interface for therapists.
- **Applied data-driven insights from user studies:** to validate system performance and iteratively improve algorithmic efficiency and operational effectiveness.

Research Intern

Korea Institute of Science and Technology

- **Heterogeneous Robot Services:** Developed and evaluated a robot-assisted system for isolation wards, incorporating telemedicine, emergency alerts, and delivery robots [2].
- **Trust in Robot Hierarchy:** Analyzed how perceived hierarchy in a robot team impacts user trust and service evaluations [3].
- **User-Control vs. Autonomy:** Studied users' preference for a reactive robot that follows verbal commands over a proactive system [4].

Graduate Researcher

Cognitive Systems Lab, Korea University

- **Research Focus:** Explored how contextual information influences human and neural network model on emotion recognition. Found contextual information significantly alters emotion perception in humans and AI systems. [5].
- **Human-Model Comparison Study:** Designed experiments to compare discrepancies between human perception and model predictions on contextual emotion recognition.
- **Contextual Emotion Model Evaluation:** Assessed the performance of a pre-trained CNN-based model on the collected In-the-Wild Dataset to analyze its effectiveness in recognizing contextual emotions.

PUBLICATIONS

- [1] N. Azizi et al., “Development of robot-assisted speech-language therapy: Co-design with speech-language pathologists,” in *Proceedings of the 16th International Conference on Social Robotics (ICSR 2024)*, Springer, 2024.
- [2] Y. Kwon et al., “Heterogeneous robot-assisted services in isolation wards: A system development and usability study,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2023, pp. 8069–8076.
- [3] S. Shin and S. S. Kwak, “Do hierarchies in a robot team impact the service evaluation by users?” In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2023, pp. 3983–3990.
- [4] S. Shin, D. Kang, and S. S. Kwak, “Is a robot trustworthy enough to delegate your control?” In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, IEEE, 2023, pp. 2415–2420.
- [5] S. Shin, D. Kim, and C. Wallraven, “Contextual modulation of affect: Comparing humans and deep neural networks,” in *Companion Publication of the 2022 International Conference on Multimodal Interaction*, 2022, pp. 127–133.

TECHNICAL SKILLS

Programming Languages: Python (Proficient), JavaScript (Proficient), C/C++ (intermediate)

Robotics and AI:

Frameworks: Robot Operating System (ROS1), NVIDIA Riva, PyTorch

Robots: QT robot, Vector

ACHIEVEMENTS

Graduate Research Studentship <i>University of Waterloo</i>	<i>Present</i>
International Doctoral Student Award <i>University of Waterloo</i>	<i>Sep 2023</i>
Provost’s Doctoral Entrance Award for Women <i>University of Waterloo</i>	<i>Sep 2023</i>

TEACHING & MENTORING EXPERIENCE

Robotics Programming Mentoring Teaching Assistant, University of Waterloo	<i>University of Waterloo, Fall 2024 - Present</i>
Fundamentals of Computational Intelligence (ECE457B) (Spring 2025)	
Programming for Performance (ECE459) (Winter 2025)	
Digital Computation (BME121) (Fall 2024)	
Algorithm Design and Analysis (ECE406) (Winter 2024)	